EE2000 #22

FPGA-Based LMS Tracking of Time-Varying ISI Channels: How Fast Can a Channel Change Before a Fixed-Point Equalizer Loses It?

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Abstract

That a least mean squares (LMS) equalizer reduces bit-error rate on a static inter-symbol interference (ISI) channel is not in question. This project asks a harder one: when the channel itself changes with time, the optimum equalizer is a moving target, and the adaptive filter must chase it. How fast can that target move before a fixed-point LMS equalizer implemented on an FPGA can no longer follow it? We build a baseband BPSK link on the Real Digital Boolean Board (Xilinx Spartan-7 XC7S50) in which the ISI channel is deliberately made time-varying under controlled and reproducible profiles — abrupt transitions and continuous ramps at selectable rates — and we measure the resulting bit-error rate, mean-square error and reacquisition time. The variation rate is a laboratory analogue of normalised Doppler, the quantity that governs equalizer tracking in mobile wireless, underwater acoustic and HF links. Three receivers operate concurrently on the identical impaired sample stream: a direct detector, an LMS equalizer trained once and then frozen, and a continuously adapting LMS equalizer. The frozen and direct paths are experimental controls that isolate the contribution of continuous adaptation. Classical theory predicts that in a nonstationary environment the total excess error decomposes into a gradient-noise term growing with the step size and a lag term falling with it, so that an optimum step size exists and scales with the rate of channel variation; we test whether that prediction survives fixed-point arithmetic, where coefficient-update underflow can halt adaptation while the channel continues to move. A coefficient word-length sweep relates tracking capability to DSP-slice and timing cost. Recorded audio is the primary mid-term payload and a grayscale image the primary final payload, both chosen so that loss of tracking is directly perceptible. Normalized LMS is identified as an extension, not as core scope. Results are reported against five explicitly separated validation levels.

1. Introduction

A communication channel with memory spreads each transmitted symbol across several symbol intervals, so that adjacent symbols overlap at the receiver. This inter-symbol interference does not diminish as transmit power rises: it scales with the signal itself, and a fixed-threshold detector therefore retains an irreducible error floor however favourable the noise conditions become. Adaptive equalization removes that floor by learning a filter which approximately inverts the channel, without the channel ever being measured explicitly [2, 9].

For a channel that does not change, this is a solved problem. The optimum equalizer $\mathbf{w}^\star$ is a fixed point, LMS [1] converges toward it, and after convergence the interesting behaviour is over. Real channels rarely oblige. Movement, changing reflections, multipath geometry and varying propagation conditions all make the channel impulse response a function of time, so the optimum equalizer becomes $\mathbf{w}^\star[n]$ — and the adaptive filter is no longer converging to a point but pursuing a trajectory. The governing question changes accordingly, from *does the filter converge* to *can the filter adapt faster than the channel changes*.

This reframing is what the project measures. A static-channel demonstration has one outcome: the equalizer works. A tracking study has a boundary — a rate of channel variation beyond which adaptation no longer keeps up — and locating that boundary under fixed-point arithmetic on real hardware is a result rather than a confirmation.

Three design decisions govern whether such a system works at all, and all three are invisible in floating-point simulation. The delay at which the equalizer is asked to produce its decision (Sec. 5.3) determines whether the problem is well posed. The precision with which coefficients are accumulated (Sec. 6.2) determines whether adaptation continues at all; in a tracking context this failure is considerably more damaging than in a static one, because the equalizer stops moving while the channel does not. The pipeline latency inserted into the coefficient-update loop to close timing changes the algorithm being implemented [15].

Both payloads are chosen so that a tracking failure is perceptible rather than merely tabulated. Because a payload is transmitted sequentially, position within it corresponds approximately to elapsed time; a channel that drifts during transmission therefore leaves a frozen-coefficient receiver progressively more mismatched, and the reconstructed image degrades from top to bottom while the adaptive reconstruction does not.

2. Real-World Applications

A channel varies whenever the transmitter, the receiver or the scatterers between them move. The resulting Doppler spread sets a coherence time T_c, the interval over which the impulse response stays approximately constant, and the quantity that decides whether an adaptive equalizer can cope is the ratio of that coherence time to the symbol period — equivalently the normalised Doppler $f_D T_s$ [8]. The variation-rate preset S of Sec. 4, expressed as symbols per coefficient increment, is a laboratory analogue of exactly that ratio, which is why sweeping it is more than a synthetic exercise.

The regime spans several orders of magnitude across deployed systems. Mobile wireless links are the canonical case, with vehicular motion decorrelating the channel within milliseconds. Underwater acoustic communication is the extreme: because the propagation speed is roughly $1500\,\mathrm{m/s}$ rather than 3×10^8, even walking-pace relative motion produces a Doppler spread that is enormous relative to the available bandwidth, and equalizer tracking rather than noise is the limiting factor. Ionospheric HF links fade as the reflecting layer shifts. At the slow end, wireline and magnetic-recording read channels drift with temperature and head position over far longer timescales, where a periodically retrained equalizer suffices.

Table 1 places these domains on a common axis. The spread is wide enough that the same algorithm serves very different receiver strategies: at the slow end a periodically retrained equalizer is adequate, which is precisely the frozen receiver R2 of Sec. 5.5, while at the fast end only continuous adaptation survives.

Two features of these systems motivate the specific axes this project measures. First, the receiver rarely knows the channel in advance, so adaptation is a requirement rather than an optimisation. Second, the equalizer runs inside a power- and area-constrained receiver — frequently an FPGA or ASIC datapath — where coefficient precision is a budgeted resource rather than a free parameter. The interaction between how fast the channel moves and how finely the coefficients are represented is therefore a practical design question, and it is the one measured here.

Table 1. Indicative variation rates across application domains and the receiver strategy each implies. Timescales are orders of magnitude for orientation, not specifications.

Domain	Timescale	Receiver strategy
Underwater acoustic	Doppler large vs. bandwidth	Continuous tracking; the limiting factor is the equalizer, not noise
Mobile wireless	Milliseconds	Continuous adaptation, pilot-assisted or decision-directed
HF skywave	Sub-second to seconds	Adaptation with frequent reacquisition
Wireline / DSL	Minutes to hours	Train at startup, retrain periodically
Magnetic recording	Per track or sector	Train per sector from a known preamble

3. Problem Statement and Objectives

Let $b[n] \in \{0, 1\}$ be the information bit sequence, mapped to BPSK symbols $x[n] = 2b[n] - 1 \in \{-1, +1\}$. For a static channel of length M,

$$r[n] = \sum_{k=0}^{M-1} h[k]\, x[n-k] + v[n]. \tag{1}$$

The project replaces $h[k]$ by $h[k, n]$, giving

$$r[n] = \sum_{k=0}^{M-1} h[k, n]\, x[n-k] + v[n], \tag{2}$$

so that the amount and structure of the ISI vary as transmission proceeds.

BPSK is used deliberately. Keeping the modulation trivial concentrates all of the difficulty in the channel, the adaptation and the arithmetic, which is where the project's questions lie. The link is an equivalent discrete-time baseband system; no RF carrier is generated, as none is required to exercise tracking.

3.1. Research question

The central question is

> *how rapidly may $h[k, n]$ vary before a fixed-point LMS equalizer can no longer track it well enough to be useful?*

Formally, we characterise

$$\mathrm{BER} = f(\mu,\, S,\, B)\big|_{\mathrm{SNR}} \tag{3}$$

where S is the channel variation rate, μ the step size and B the coefficient word length, evaluated at a fixed signal-to-noise ratio (Sec. 10), and identify the boundary in S beyond which the adaptive receiver ceases to outperform its controls.

3.2. Objectives

1. define controlled, reproducible time-varying channel profiles with explicitly specified coefficient trajectories and selectable variation rates (Sec. 4);
2. implement an LMS equalizer in fixed-point arithmetic with decision delay, training sequence and coefficient precision treated as explicit design parameters;
3. **measure tracking capability** as bit-error rate, mean-square error and reacquisition time against channel variation rate, for a direct detector, a frozen equalizer and a continuously adapting equalizer operating on one impaired stream;
4. **test the step-size prediction** that an optimum μ exists for each variation rate and scales with it (Sec. 5.2), and determine whether that relationship survives fixed-point implementation;
5. **relate precision to tracking** by sweeping coefficient word length and recording the maximum trackable variation rate against DSP-slice, LUT and timing cost.

Objectives 3–5 are the contributions. Objectives 1 and 2 are the apparatus required to obtain them.

4. Time-Varying Channel Model

The channel of (2) is realised as a short FIR filter whose coefficients are driven by a variation controller. Two trajectory families are specified, and in both cases the coefficient evolution is stated explicitly rather than left to a qualitative description.

4.1. Abrupt transition

The channel holds $\mathbf{h}_A$ until a known symbol index N_C, then switches to $\mathbf{h}_B$, for example

$$[1,\ 0.3,\ 0.1] \longrightarrow [1,\ 0.6,\ 0.1]. \tag{4}$$

This isolates *reacquisition*: the error rises, the coefficients move, and the filter reconverges. The measured quantity is the reacquisition time N_{rec}, defined as the number of symbols after N_C required for the running mean-square error to return within a stated factor of its pre-transition steady-state value. The factor is fixed before measurement and reported with the results.

4.2. Continuous variation

The channel is $\mathbf{h}[n] = [1,\ a[n],\ 0.1]$ with $a[n]$ following a periodic triangular trajectory between $a_{\min}$ and $a_{\max}$ (provisionally 0.20 and 0.60), incremented by a fixed step every S symbols. A sinusoidal profile $a[n] = a_0 + A\sin(2\pi f_c n)$ is the more conventional model, but a triangular ramp is generated by an up/down counter and requires no sine hardware, which matters for the RTL stage. The floating-point model will confirm that the two give comparable tracking behaviour at matched average rates of change.

Table 2. Channel variation rate presets. Values are provisional and will be fixed after floating-point simulation so that the set brackets the observed tracking boundary.

Preset	Behaviour	Symbols/step
S_0	Static (control)	∞
S_1	Very slow drift	1000
S_2	Slow drift	200
S_3	Moderate drift	50
S_4	Fast drift	10

Figure 2 shows both families. The variation rate is parameterised by the preset S of Table 2; smaller S means a faster-changing channel. Final preset values will be fixed after simulation, since they must bracket the tracking boundary rather than sit entirely on one side of it.

4.3. Noise generation

Pseudo-random noise is added after the channel filter. An approximately Gaussian sample is formed by summing several independent uniform variates. Independence is essential: bit-slices or shifted taps drawn from a single LFSR are correlated and produce a degenerate distribution that can flatter the measured bit-error rate. Separate LFSRs with distinct primitive polynomials and seeds will therefore be used.

Validation will be by mean, variance, autocorrelation, saturation behaviour, seed reproducibility and effect on bit-error rate, not by histogram shape alone. One limitation will be stated with results: the sum of N uniform variates has bounded support, so excursions beyond $\pm\sqrt{3N}\,\sigma$ cannot occur — $\pm 6\sigma$ for the common choice $N = 12$ — and measurements will accordingly be quoted only down to approximately 10^{-4}.

Every comparison between receivers, step sizes or word lengths uses identical transmitted data, identical channel trajectory and identical noise realisation. This is enforced by seeding rather than assumed.

5. LMS Tracking

5.1. Algorithm

For an L-tap equalizer with input and coefficient vectors

$$\mathbf{u}[n] = [\,r[n],\ r[n-1],\ \ldots,\ r[n-L+1]\,]^T, \tag{5}$$
$$\mathbf{w}[n] = [\,w_0[n],\ \ldots,\ w_{L-1}[n]\,]^T, \tag{6}$$

the output is $y[n] = \mathbf{w}^T[n]\mathbf{u}[n]$ and the update is

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu\, e[n]\, \mathbf{u}[n]. \tag{7}$$

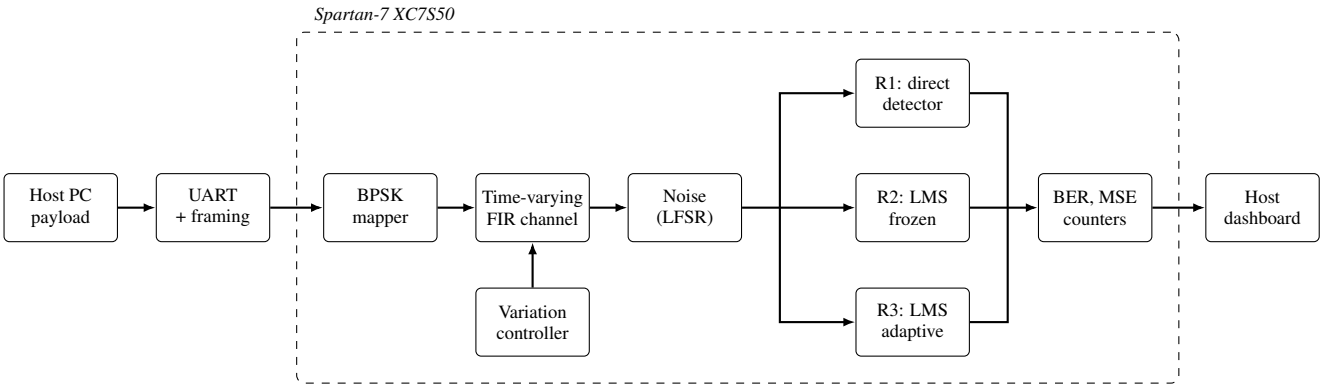
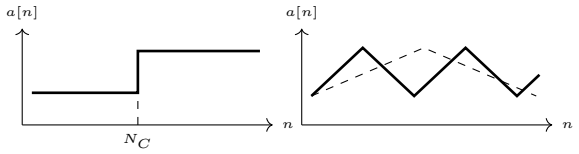


Figure 1. Proposed system. A variation controller drives the channel coefficients along a specified trajectory at a selectable rate. One channel realisation and one noise seed feed three concurrent receivers, so differences between them are attributable to the adaptation rule alone. R2 requires only a fixed FIR after training; only R3 carries coefficient-update hardware.



(a) abrupt transition (b) triangular ramps

Figure 2. Specified channel coefficient trajectories. In (b) the solid line is a fast preset and the dashed line a slow preset of the same amplitude; the variation rate S is the number of symbols between coefficient increments. Schematic; rates are set in Table 2 and amplitudes in Sec. 4.

Mean-square convergence requires

$$0 < \mu < \frac{2}{\mathrm{tr}(\mathbf{R})} = \frac{2}{L\,E[r^2[n]]}, \tag{8}$$

where $\mathbf{R} = E[\mathbf{u}\mathbf{u}^T]$. In hardware μ is constrained to a power of two so that the scaling in (7) is a shift rather than a multiplier.

5.2. The tracking–accuracy trade-off

Write $J_{\min}$ for the irreducible mean-square error and define the misadjustment $\mathcal{M} = \xi_{\mathrm{ex}}/J_{\min}$ as excess error normalised by it. For a stationary channel the misadjustment is dominated by gradient noise and grows roughly as $\mu\,\mathrm{tr}(\mathbf{R})/2$ [5], so smaller μ is simply better once convergence is complete.

A nonstationary channel breaks that monotonicity. Because $\mathbf{w}[n]$ trails the moving optimum $\mathbf{w}^\star[n]$, a second term appears, and it grows as μ falls. Modelling the optimum trajectory as a random walk with per-sample increment covariance $\mathbf{Q}$, the classical result [3, 5, 6] is

$$\mathcal{M} \approx \underbrace{\frac{\mu\,\mathrm{tr}(\mathbf{R})}{2}}_{\text{gradient noise}} + \underbrace{\frac{\mathrm{tr}(\mathbf{Q})}{2\mu J_{\min}}}_{\text{lag}}, \tag{9}$$

which is minimised at

$$\mu_{\mathrm{opt}} \approx \sqrt{\frac{\mathrm{tr}(\mathbf{Q})}{J_{\min}\,\mathrm{tr}(\mathbf{R})}}. \tag{10}$$

Exact constants depend on the nonstationarity model assumed; the structural content — one term rising with μ, one falling, hence an interior optimum — does not.

Equation (10) is the theoretical backbone of the project, and it is falsifiable. It predicts that a faster-varying channel — larger $\mathrm{tr}(\mathbf{Q})$ — calls for a larger step size, and specifically that μ_{opt} should scale as the square root of the variation rate. The step-size sweep of Sec. 10 tests that scaling directly.

Two caveats will be stated rather than glossed. First, the trajectories of Sec. 4 are deterministic ramps and steps, not random walks, so (10) is an approximation whose accuracy is itself worth measuring. Second, (9) assumes infinite precision; in fixed point the lag term acquires an additional contribution once coefficient updates fall below the quantisation floor (Sec. 6.2), and the measured optimum may depart from the prediction. Establishing whether it does is one of the more interesting outcomes available to the project.

5.3. Decision delay

The error signal carries an explicit delay,

$$e[n] = x[n - \Delta] - y[n], \tag{11}$$

where Δ is the equalizer decision delay. The delay is required for the problem to be well posed, not merely convenient.

The cascade of channel and equalizer is $c = h * w$, with design target $c[n] \approx \delta[n-\Delta]$: a clean impulse at *some* delay, not necessarily at zero. Setting $\Delta = 0$ would require w to be the exact inverse of h, which is in general infinite in extent, and which for a non-minimum-phase h is anti-causal and so unrealisable by a causal FIR filter of any length.

EE2000 #22

The same conclusion follows by counting samples, as illustrated in Fig. 3. For $h = [0.5,\ 1.0]$, the symbol $x[n]$ contributes to $r[n]$ with weight 0.5 and to $r[n+1]$ with weight 1.0. At time n the window spans $r[n]$ back to $r[n-L+1]$ and so holds only $0.5^2/(0.5^2 + 1.0^2) = 20\%$ of that symbol's received energy, whereas $x[n-1]$ has both contributions inside. Requesting $x[n]$ asks the filter to reconstruct a symbol whose dominant evidence has not yet arrived; LMS settles on the best available compromise and the mean-square error plateaus above the achievable floor with no overflow or instability to signal the fault.

Δ will be selected by sweeping $0 \le \Delta \le L+M-2$ in the floating-point model and recording steady-state mean-square error, which exhibits a distinct minimum. Because at least one channel preset is non-minimum phase — $h = [0.5, 1.0]$ and $h = [1.0, 0.5]$ have identical magnitude responses but zeros at $z = -2$ and $z = -0.5$, and very different equalization difficulty — this sweep is necessary rather than precautionary. A further question specific to this project is whether the best Δ remains constant as the channel varies, or whether the trajectory should be constrained so that it does. In hardware Δ is a shift register on the desired-symbol path and costs Δ flip-flops.

One consequence propagates to the host: recovered bits are delayed by Δ, and each receiver carries its own offset, the direct detector's being the index of the dominant channel tap. Comparing without the correct per-path offset yields a measured bit-error rate near 0.5 from a perfectly functioning equalizer. The host will cross-correlate recovered against transmitted bits and report the peak lag as a check against the predicted offset.

5.4. Training sequence

LMS converges toward $\mathbf{R}^{-1}\mathbf{p}$. If the training sequence is periodic, $\mathbf{R}$ is rank deficient: only part of the spectrum is excited, the equalizer is unconstrained in the remaining subspace, and it may converge to a filter that inverts the channel only at the excited frequencies. Adaptation appears to succeed and then fails on payload data. This is the persistent excitation condition, and an alternating ± 1 pattern is its worst case, exciting a single frequency.

Training therefore uses a maximal-length LFSR sequence, whose autocorrelation is close to an impulse, giving $\mathbf{R} \approx \sigma^2\mathbf{I}$. This satisfies persistent excitation and simultaneously minimises the eigenvalue spread $\lambda_{\max}/\lambda_{\min}$ governing convergence rate [5] — of direct consequence here, since convergence rate sets the reacquisition time N_{rec}. The polynomial and length will be reported. Hardware cost is one shift register and an XOR gate.

5.5. Three receivers

The impaired sequence $r[n]$ drives three receivers sharing one tapped delay line:

R1, direct. Threshold detection on $r[n]$, no equalization. The uncorrected error floor.

R2, frozen. LMS adapts during training; at $n = N_T$ the coefficients $\mathbf{w}_F = \mathbf{w}[N_T]$ are stored and held. This represents an equalizer matched to the channel as it was, unable to follow subsequent change. After training it is a fixed FIR filter and requires no update hardware [9].

R3, adaptive. Adaptation continues throughout the payload, tracking $h[k, n]$.

R1 and R2 are experimental controls, not independent contributions: R1 isolates the value of equalization and R2 isolates the value of *continued* equalization. The expected ordering — R3 degrading more slowly than R2, which degrades faster than it began — is a hypothesis to be measured, and conditions under which R2 matches or beats R3 (very slow variation, where R3 pays gradient noise for tracking it does not need) will be reported rather than omitted.

5.6. Adaptation source during payload

After training, R3 requires an error signal without a known reference. Three modes are distinguished, and any reported result states which was used:

Decision-directed. $\hat{d}[n] = \text{sign}(y[n])$ substitutes for the reference in (11) [11, 12]. Realistic, but divergent once enough decisions are wrong; the transition into this mode is gated on a measured error threshold rather than a fixed symbol count, and the threshold is reported.

Pilot-assisted. Known symbols are inserted periodically, bounding error accumulation at a stated overhead in payload rate.

Supervised evaluation. Because the transmitter and channel are internal to the test environment, the true symbol is available. This is used only for controlled experiments that isolate tracking from decision errors, and is labelled as supervised evaluation, never presented as an independently realisable receiver.

5.7. Detection and metrics

Each path applies $\hat{b}[n] = 1$ for $y[n] \ge 0$ and 0 otherwise, with $\text{BER} = N_{\text{error}}/N_{\text{total}}$ computed separately. Beyond bit-error rate, tracking is characterised by the instantaneous squared error $e^2[n]$ and its running mean, the coefficient trajectories $w_k[n]$ compared against the known channel trajectory, the reacquisition time N_{rec}, and the maximum trackable variation rate $S^\star$. The last requires an operational definition, fixed before measurement: the fastest preset at which R3 maintains a bit-error rate within a stated factor of its static-channel value.

EE2000
#22

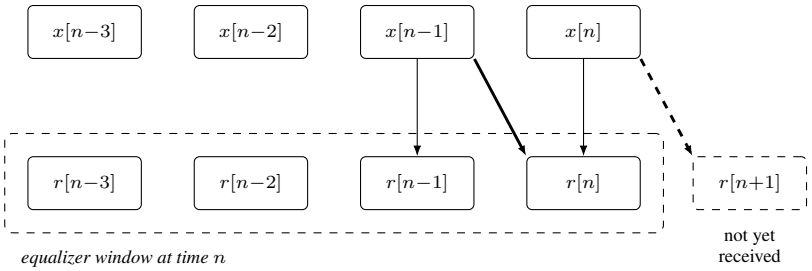


Figure 3. Why the decision delay is necessary, for $h = [0.5,\ 1.0]$; line thickness indicates tap weight. Each transmitted symbol contributes to two received samples. At time n the equalizer observes $r[n]$ and earlier, so $x[n-1]$ is fully represented inside the window while the dominant contribution of $x[n]$ falls in $r[n+1]$, which has not arrived. Setting $\Delta = 0$ therefore asks the filter to reconstruct a symbol it has largely not yet observed.

6. Fixed-Point Implementation

6.1. Datapath widths

Signal width, coefficient width, accumulator width, fractional-bit allocation, rounding and saturation are each chosen explicitly. Multiplication widens the intermediate result and accumulation over L taps widens it further, so internal accumulators carry headroom and are rescaled once at the output. Saturation is used in preference to two's-complement wraparound, since a wrapped value inverts sign and can drive the adaptation loop unstable rather than merely clipping it.

6.2. Coefficient update underflow

The update path fails at the opposite end of the range from the output path, and it fails silently. If w is stored as Q1.14, its least significant bit is $2^{-14} \approx 6.1 \times 10^{-5}$. With $\mu = 2^{-12}$, $|e| \approx 0.05$ and $|u| \approx 1$, the update term $\mu e[n]u[n] \approx 1.2 \times 10^{-5}$ falls below that bit and rounds to zero. Adaptation stops. No overflow occurs and no flag is raised.

On a static channel this merely halts convergence early, leaving a filter that works but works poorly. On a time-varying channel the consequence is qualitatively worse: the equalizer ceases to move while the channel continues to, so the mismatch grows without bound and R3 degenerates into R2. A fixed-point tracking system can therefore fail in a way that a fixed-point static system cannot, which makes this the single most important arithmetic decision in the project.

The coefficients will accordingly be carried in an extended-precision accumulator, provisionally Q1.24, from which the most significant bits feed the multiplier. Coefficient word length is thus two parameters — accumulation width and multiply width — and only the latter enters DSP-slice cost. Detection is by comparing fixed-point against floating-point coefficient trajectories while the channel moves; the divergence is unmistakable. Related stability concerns for long-running fixed-point equalizers are treated in [16].

6.3. Delayed coefficient adaptation

The path $\mathbf{w} \to y \to e \to \mathbf{w}$ is a combinational feedback loop and cannot be pipelined without altering the algorithm: inserting D registers yields delayed LMS, whose convergence requires a tightened bound on μ [15]. At the initial UART-rate target of approximately 10^4 symbols/s against a 100 MHz clock, roughly 10^4 clock cycles are available per symbol, so $D = 0$ is achievable. The pipelined variant becomes relevant only for a higher-rate mode. Where pipelining is used, D and the resulting bound are stated explicitly.

6.4. Word length and tracking

Coefficient width is a Verilog parameter, synthesised at 8, 12 and 16 bits. For each, the design records DSP48E1 count, LUT and flip-flop utilisation, maximum frequency and timing slack, together with bit-error rate, reacquisition time and — the quantity specific to this project — the maximum trackable variation rate $S^\star$. The governing question is how much precision is required to sustain tracking at a given rate, and the expected mechanism linking the two is the underflow of Sec. 6.2, which gives the resulting curve a physical explanation rather than an empirical one. Word-length selection for linear DSP datapaths is treated systematically in [17].

6.5. Resource budget

Table 3 estimates the arithmetic required, to confirm before RTL work that three concurrent receivers fit. A fully parallel L-tap LMS equalizer at one sample per clock needs L multipliers for filtering and L for the update; with μ a power of two the scaling costs none. R2, being a fixed FIR after training, needs only L. These are architectural estimates at validation level L1, not synthesis results; their purpose is to establish feasibility and identify DSP slices as the binding constraint.

Table 3. Design-time resource estimate at $L = 16$, $M = 3$. Estimates only; actual utilisation is measured at synthesis.

Block	DSP48E1	BRAM
Time-varying channel FIR ($M = 3$)	3	–
Variation controller, noise LFSRs	0	–
R1 direct detector	0	–
R2 frozen FIR (no update logic)	16	–
R3 adaptive LMS (filter + update)	32	–
Payload and recovered buffers	–	≈64 kB
Estimated total	**51**	**≈64 kB**
Available on XC7S50	**120**	**337 kB**
Estimated utilisation	**43%**	**19%**

7. Payload Modes

Payloads are ordered by role rather than by difficulty. Text supports bring-up only. Recorded audio is the primary mid-term demonstration. A grayscale image is the primary final demonstration. Live capture is a gated extension outside the core (Sec. 11.2).

7.1. Text

Characters are serialised to bits and reconstructed from each receiver output. Corruption is immediately legible, which makes this the appropriate bring-up payload; it is not used as a result.

7.2. Recorded audio

A short mono clip is quantised, converted to bits, passed through the link and reconstructed once per receiver, producing `original`, `direct`, `frozen` and `adaptive` files. Tracking failure becomes audible.

Real-time operation is not required here, and recorded audio is preferred for the core work for a reason that is not about throughput: it is *replayable*. Every experiment in Sec. 10 demands identical transmitted data, channel trajectory and noise seed while S, μ and B are swept. A recorded clip satisfies that; a live microphone input cannot, by construction. The workflow is therefore record, transmit, reconstruct, play, stated plainly rather than presented as streaming. Live capture is treated as a demonstration extension in Sec. 11.2, gated on completion of the core.

This is communication-channel equalization, not acoustic noise cancellation. The latter requires a reference input correlated with the unwanted acoustic noise and is a different architecture; no claim of acoustic noise removal is made.

7.3. Grayscale image

The final visual demonstration uses a 128×72 image at 8 bits/pixel, giving 73,728 payload bits. Buffered on-chip this occupies a small fraction of available block RAM, so subdivision into independently processed frames is unnecessary. Two levels of framing are distinguished:

- *Transport framing* is retained; without headers, length fields and sequence numbers a single lost UART byte shifts all subsequent data and shears the reconstruction.
- *Adaptation framing is removed.* The equalizer trains once and retains coefficients throughout. Per-frame retraining would place 500–2000 pre-convergence symbols at each frame boundary, producing periodic horizontal bands indistinguishable from a fault — and, more seriously here, would conceal the tracking behaviour the experiment exists to observe. Retraining is placed under pushbutton control.

Sequential transmission makes position within the image correspond approximately to elapsed time. A channel drifting during transmission therefore renders tracking ability spatially: R2 degrades progressively down the image while R3 does not, until at sufficiently fast variation R3 also fails. Reported metrics include bit-error rate, incorrect pixel count, mean absolute pixel error, and a per-bit-plane error breakdown, since an error in the most significant bit shifts a pixel by 128 grey levels while one in the least significant bit shifts it by one.

Table 4. Planned Boolean Board interaction.

Resource	Planned use
Switches	Channel preset, variation on/off, variation-rate preset S, noise level, adaptation enable, payload mode
Pushbuttons	Reset, start, retrain, freeze coefficients, clear counters
LEDs / RGB	Blue: training. Green: tracking. Red: tracking lost or high error
7-segment	Error count, variation-rate preset, mode, status
Potentiometer	Continuous noise-variance control (fallback: switches)
USB-UART	Commands, payload, returned data, diagnostics

8. Host–FPGA Interface and Board Interaction

The host performs media conversion, packet preparation, UART transfer, reassembly, plotting and playback; the FPGA performs mapping, channel variation, noise, LMS adaptation, detection, counting and board status. Table 4 gives the planned mapping. Making the variation rate a front-panel control is what turns the central research question into something an examiner can operate directly: the rate is increased until the adaptive receiver visibly loses tracking.

9. Development and Verification

Work proceeds so that faults are isolated rather than discovered after integration.

P1 Floating-point static LMS. Fixes Δ by sweep (Sec. 5.3), verifies the LFSR training sequence (Sec. 5.4), establishes baseline μ and L.

P2 Abrupt channel transition; measures N_{rec}.

P3 Continuously varying channel.

P4 Variation-rate and step-size sweeps; tests the prediction (10).

P5 Recorded-audio experiment end to end in software.

P6 Fixed-point model including the extended-precision accumulator (Sec. 6.2); reference vectors exported.

P7 RTL for the LMS core and the channel variation controller.

P8 RTL verification against P6 reference vectors.

P9 Vivado synthesis and implementation at each word length.

P10 Board bring-up, in order: status LED, UART echo, mapper, channel, detector, equalizer, known bit pattern, text, audio, image.

P11 Optional: NLMS (Sec. 11.1).

P12 Optional: live audio (Sec. 11.2).

P1–P6 are software and constitute the mid-term target; P7–P10 are hardware and constitute the final target. The issues of Sec. 5.3, 5.4 and 6.2 are all resolved within P1–P6, before RTL entry, because each is invisible in floating point and each produces symptoms resembling RTL faults.

10. Experimental Plan

Unless stated otherwise, every tracking experiment is run at a single fixed operating signal-to-noise ratio, chosen in P1 so that the static-channel bit-error rate is low enough that degradation is attributable to tracking loss rather than to noise. Holding noise fixed while S, μ and B vary is what makes the comparisons interpretable; the noise level is reported with every result, and a separate sweep establishes sensitivity to it.

10.1. Baseline validation

Before any tracking result is claimed, bit-error rate is measured against signal-to-noise ratio on a *static* channel, across approximately twelve noise levels with at least 10^6 bits per point, and plotted against the theoretical BPSK-in-AWGN bound.

This measurement validates the apparatus rather than testing a hypothesis, and it does so economically: agreement between the flat-channel points and theory simultaneously confirms correct noise scaling and distribution (Sec. 4.3), correct detection threshold, correct signal-to-noise-ratio calibration, and correct per-path bit alignment (Sec. 5.3). Departure from theory, and the shape of that departure, localises which of the four is at fault. The static ISI channel is expected to show an error floor that the equalizer substantially removes. Only 10^6-bit error counts return over UART, so the measurement is cheap to run at every word length.

10.2. Tracking experiments

Principal result. Bit-error rate against channel variation rate S, for R1, R2 and R3 on identical data, channel and noise. The expected qualitative behaviour — R1 flat and poor, R2 good at S_0 and degrading as S increases, R3 holding longer before also failing — is a hypothesis, and the location of R3's failure is the measurement.

Step-size study. The above repeated for several μ, reported as a surface or heatmap over (μ, S), and tested against the $\sqrt{\cdot}$ scaling predicted by (10).

Reacquisition. Mean-square error against time around an abrupt transition, and N_{rec} against μ.

Trajectories. Coefficients $w_k[n]$ overlaid on the known channel trajectory, showing the filter following or failing to follow.

Precision. Bit-error rate, N_{rec} and $S^\star$ against coefficient word length, with the corresponding hardware cost.

Payload. Original against direct, frozen and adaptive reconstructions, for both audio and image.

Bit-error-rate experiments are repeated across independent seeds and reported as averages. Every reported quantity is labelled with the validation level that produced it.

11. Optional Extensions

Two extensions are identified. Both are gated on completion of the core: neither is started until P1–P10 are complete and both measurement studies of Sec. 10 are reported. NLMS is ranked first because it deepens the adaptive-filtering question the project exists to answer; live audio is ranked second because it deepens the input/output system instead. If neither is reached, nothing in the core result is diminished.

11.1. Normalized LMS

NLMS is an extension, not core scope, and the project stands complete without it. The update

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \frac{\mu}{\epsilon + \|\mathbf{u}[n]\|^2}\, e[n]\, \mathbf{u}[n] \qquad (12)$$

normalises the step by input energy, giving an effective step size that adjusts as received signal statistics change [13, 14].

EE2000 #22

The question it addresses is specific: does normalisation extend the trackable variation range, and by enough to justify its hardware?

That hardware is the obstacle. LMS with $\mu = 2^{-p}$ needs only a shift. NLMS additionally requires $P[n] = \epsilon + \sum_k u_k^2[n]$ and then $1/P[n]$ — input energy accumulation and a reciprocal, by lookup table, piecewise approximation, divider IP or iterative method, each with its own fixed-point analysis. This is why NLMS is a genuine hardware extension rather than another receiver path, and why the development order is floating-point LMS, time-varying LMS, fixed-point LMS, FPGA LMS, and only then NLMS. No assumption is made that NLMS will outperform LMS; the comparison, if reached, will be measured on identical data, channel and noise.

11.2. Live audio

If the core is complete and NLMS either finished or abandoned, live audio becomes the final extension. It converts the project from a batch experiment into a continuously running receiver, and it requires one item of external hardware: a Pmod microphone module on one of the board's expansion connectors.

The naive architecture does not close. Returning three recovered audio streams to the host costs $3 \times 64 = 192$ kbit/s, plus 64 kbit/s forward, against roughly 92 kbit/s of usable throughput on a 115200-baud 8N1 link — a factor of 2.8 deficit. Raising the baud rate is one answer, but a better one uses hardware already present: route the selected recovered stream to the board's PWM audio output and listen to it directly, with a switch choosing which of R1, R2 or R3 is audible. No audio then crosses the UART at all, which carries only counters and status, and the bottleneck disappears rather than being widened. Being able to switch between the unequalized, frozen and adapting receivers *while listening* is also a more direct demonstration than three files played in sequence.

Two feasibility checks support this route. Sampling at 8 kHz places the Nyquist limit at 4 kHz, inside the board's 50 Hz–5 kHz audio passband and comfortably above the 300–3400 Hz band that carries speech intelligibility. At 8 bits/sample the symbol rate becomes 64 ksym/s, leaving roughly 1560 clock cycles per symbol at 100 MHz — still ample for the coefficient-update loop to settle within one symbol, so the delayed-LMS variant of Sec. 6.3 remains unnecessary.

What the extension costs is a microphone interface, a PWM audio driver, and a streaming controller that must sustain throughput without the buffering slack the batch modes enjoy. What it cannot do is produce a measurement. A live acoustic input is unrepeatable, so no bit-error rate, tracking boundary or word-length comparison may be drawn from it; it is a demonstration, and any result reported from it would violate the fair-comparison condition of Sec. 4.3. That limitation, not the data rate, is the reason it is ranked last and excluded from the core.

12. Related Literature

12.1. Origins and theory

LMS originates with Widrow and Hoff [1] and the first automatic equalizer with Lucky [2]. Widrow and Stearns [4] give the foundational treatment; Haykin [5] develops convergence, eigenvalue spread and the misadjustment–step-size relation underlying Sec. 5.2.

12.2. Tracking and nonstationary environments

This is the cluster most directly relevant to the project. Widrow *et al.* [3] analyse LMS learning characteristics in stationary and nonstationary environments and establish the decomposition of excess error into gradient-noise and lag components that yields (9) and (10). Macchi [6] treats tracking specifically in a transmission context, and Eleftheriou and Falconer [7] compare tracking performance across algorithm families, providing the framework in which LMS tracking results are conventionally reported. Jakes [8] supplies the standard statistical models of time-varying channels against which the deliberately simplified deterministic trajectories of Sec. 4 should be positioned.

12.3. Channel equalization

Qureshi [9] remains the standard survey, covering equalizer structures, training strategies, equalizer delay, and the difficulty presented by non-minimum-phase channels. Proakis and Salehi [10] provide the communication-system framing and the ISI model of (2).

12.4. Decision-directed and normalized adaptation

Sato [11] and Godard [12] established adaptation without a continuous training reference and characterise convergence and divergence conditions, informing Sec. 5.6. Nagumo and Noda [13] introduced normalised adaptation and Slock [14] analyses LMS and NLMS convergence, relevant to Sec. 11.1.

12.5. FPGA implementation and fixed-point design

Long, Ling and Proakis [15] analyse LMS with delayed coefficient adaptation and derive the modified step-size bound, underpinning Sec. 6.3. Hanson [18] surveys FPGA architectures for LMS and NLMS with their resource–throughput trade-offs. The DSP48E1 slice is documented in [19]. Constantinides *et al.* [17] address word-length selection, and Gitlin *et al.* [16] analyse coefficient drift and stability in digitally implemented equalizers — the context for Sec. 6.2.

12.6. Position of this work

LMS tracking theory is established, and FPGA implementations of LMS are documented, but the two literatures rarely meet. Tracking analyses generally assume infinite precision; FPGA implementation studies generally assume a static channel and report convergence rather than tracking. What is rarely reported in a single controlled study is the intersection: how far finite coefficient precision restricts the rate of channel variation that an adaptive equalizer can follow, measured on one device against identical channel trajectories, with hardware cost recorded alongside. That intersection is what this project contributes. MathWorks [20] is used only as an independent behavioural cross-check on the Python model; platform details follow [21, 22].

13. Demonstration

The board is reset and a payload selected. The examiner chooses a channel preset, a variation-rate preset and a noise level from the board controls. The FPGA maps the payload to BPSK, applies the time-varying channel and noise, trains on the LFSR sequence, copies the trained coefficients into the frozen bank, and processes the identical impaired stream through all three receivers. Recovered payloads, bit-error rates, mean-square error, the current channel state and the live LMS coefficients return to the host.

The examiner then increases the variation rate step by step. At low rates all three reconstructions behave as at S_0. As the rate rises the frozen reconstruction degrades progressively across the payload while the adaptive one holds. At some preset the adaptive reconstruction also begins to fail, and the RGB indicator turns from green to red. That transition is the central result of the project, made directly observable.

Limits are demonstrated as deliberately as successes: a very slow variation under which frozen and adaptive are indistinguishable, and a very fast one under which neither survives. Reporting the operating range is more informative than presenting a single favourable case.

14. Scope and Validation Status

The mandatory core is LMS tracking of a controlled time-varying ISI channel, comprising the floating-point model, the fixed-point model, the tracking experiments, the FPGA implementation and the hardware measurements. The direct and frozen receivers are experimental baselines. Recorded audio is the mid-term demonstration and the image the final visual demonstration. NLMS is an advanced extension and live audio an optional hardware extension, both gated on core completion (Sec. 11); unrelated application domains are excluded by design, since depth here comes from tracking and implementation rather than from payload variety.

Results are reported against five levels: (L1) floating-point simulation, (L2) fixed-point simulation, (L3) RTL simulation, (L4) Vivado synthesis and implementation, (L5) physical-board measurement. No quantity obtained at one level is presented as though obtained at a higher one.

Table 5. Principal risks and contingencies. Each fallback is chosen to preserve a reportable result rather than to rescue the demonstration.

Risk	Contingency
Preset rates all sit on one side of the tracking boundary	Rates are fixed after P4 simulation specifically to bracket it
Fixed-point underflow halts tracking	Extended-precision accumulator (Sec. 6.2); widen and report the required width as a result
Decision-directed mode diverges	Report the divergence threshold as a measured result; use pilot or supervised mode for controlled tracking experiments
Update loop fails timing	Divided clock; or pipeline to delayed LMS with stated D and tightened μ
Board bring-up overruns	Report L1–L4 in full and state hardware status honestly rather than substituting simulation
Extensions not reached	NLMS and live audio are both gated on core completion; LMS tracking results stand alone without either

Table 5 lists the failure modes considered most likely. The project succeeds if a tracking boundary $S^\star$ is located and shown to depend on μ and on coefficient word length in a measurable way, with reacquisition time characterised around abrupt transitions. A result in which continuous adaptation fails to help under some condition is a valid outcome and will be reported; only an inability to measure constitutes failure.

A prior software experiment, using one 128×72 test image under one static channel and noise setting, recorded 4,911 bit errors (6.661%) without equalization and 380 (0.515%) with it. These are L1 results on a static channel, quoted only as evidence that the apparatus functions, and they are not tracking results.

15. AI Usage Disclosure

Generative AI tools assisted with report organisation and wording, with identifying relevant literature, with reviewing the proposed design for known failure modes, and with planning the experimental structure. Several specific issues documented in Sec. 5.3, 5.4 and 6.2 were identified during AI-assisted review of an earlier draft and subsequently verified by the team against the cited literature.

All technical content has been checked by the team. No

simulation output, RTL behaviour, synthesis result or physical measurement in this or any subsequent report is AI-generated; all such results are produced and verified independently. AI-generated content is not treated as experimental evidence.

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